ढांचागत नक्शा (STRUCTURAL BLUEPRINT)

G+1 फार्म स्टोरेज + आवास (Farm Storage + Residence)

मथिनपुर उर्फ़ नईमतुल्ला नगर, मुरादाबाद, उत्तर प्रदेश

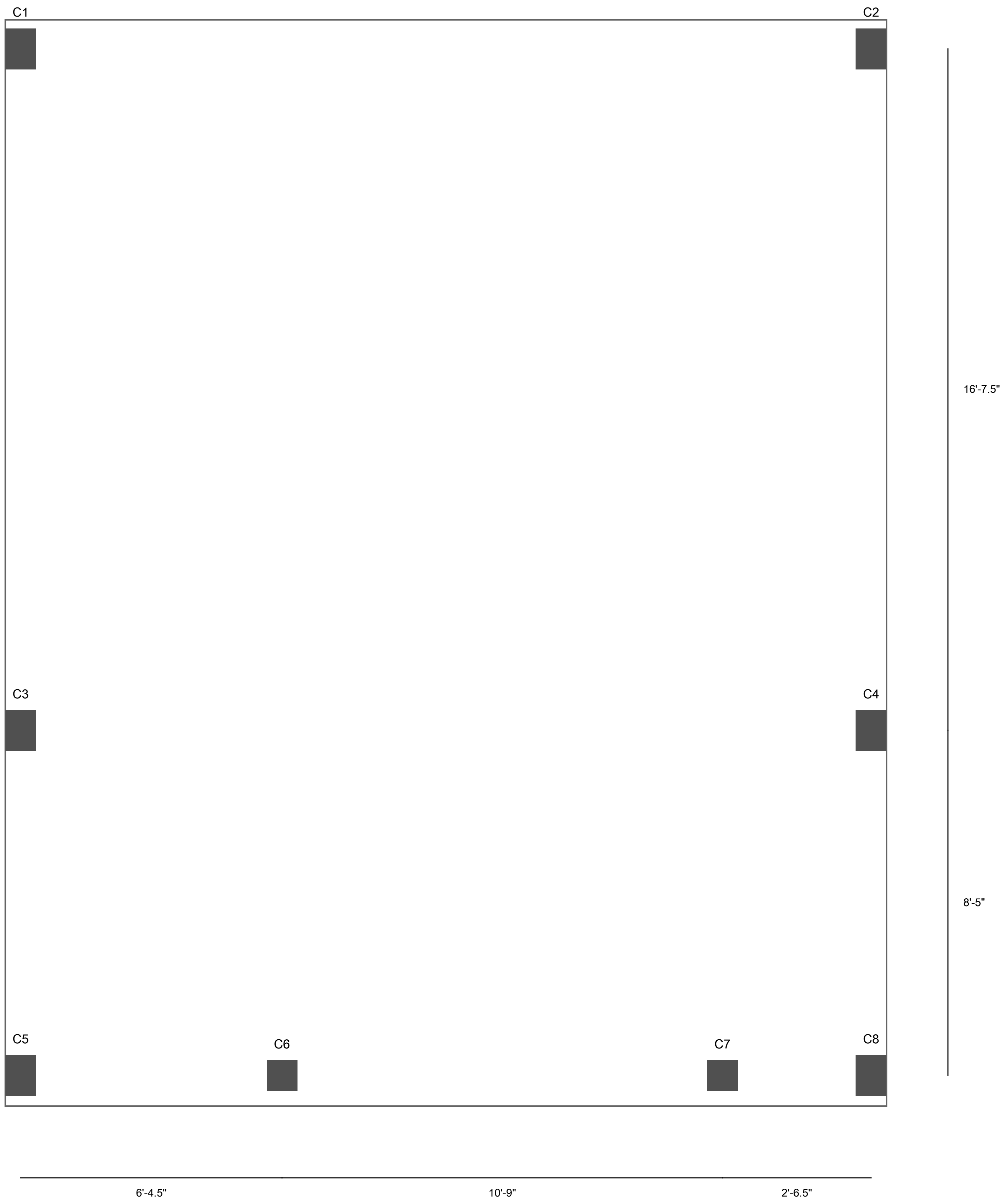
भवन का प्रकार (Type)	G+1 RCC फ्रेम (गोदाम + मकान)
प्लॉट (Plot)	20 फुट x 25 फुट अंदरूनी (21.5 x 26.5 बाहरी)
ग्राउंड फ्लोर (GF)	कृषि भंडारण (गोदाम) - 10 फुट शटर
पहली मंजिल (1F)	1BHK आवास (कमरा + रसोई + शौचालय)
ढांचा (Frame)	RCC मोमेंट फ्रेम, 8 पिलर
प्लिंथ ऊंचाई	3 फुट (जमीन से ऊपर)
मंजिल की ऊंचाई	12 फुट प्रति मंजिल
कुल ऊंचाई	28 फुट छत तक, 31 फुट पैरापेट
आगे का छज्जा	6 फुट कैंटीलीवर (बिना सपोर्ट)
भूकंप ज़ोन (Seismic)	Zone IV (मुरादाबाद) - Z = 0.24
कंक्रीट (Concrete)	M20 ग्रेड (1:1.5:3)
सरिया (Steel)	Fe500 TMT (TATA/Kamdhenu/JSW 500D)
सीमेंट	ACC / Ambuja / Ultratech OPC-43
सरिया (TMT)	TATA Tiscon / Kamdhenu 500D / JSW NeoSteel
रोड़ी (Aggregate)	20mm - Haridwar/Roorkee
बालू (Sand)	Zone-II नदी रेत - गंगा/रामगंगा
ईंट (Brick)	पक्की ईंट - स्थानीय भट्टा (1st class)

दिनांक: जून 2026 | Drawing No: STR-001-HINDI

नोट: यह नक्शा ठेकेदार/मिस्त्री के लिए है। सभी माप फुट/इंच में।

SHEET 2: COLUMN GRID PLAN (As-Built Positions)

Plan View - Looking Down. Front = Bottom. All dimensions center-to-center.



External: 21'-6" x 26'-6" | Internal: 20'-0" x 25'-0"

COLUMN SIZES:

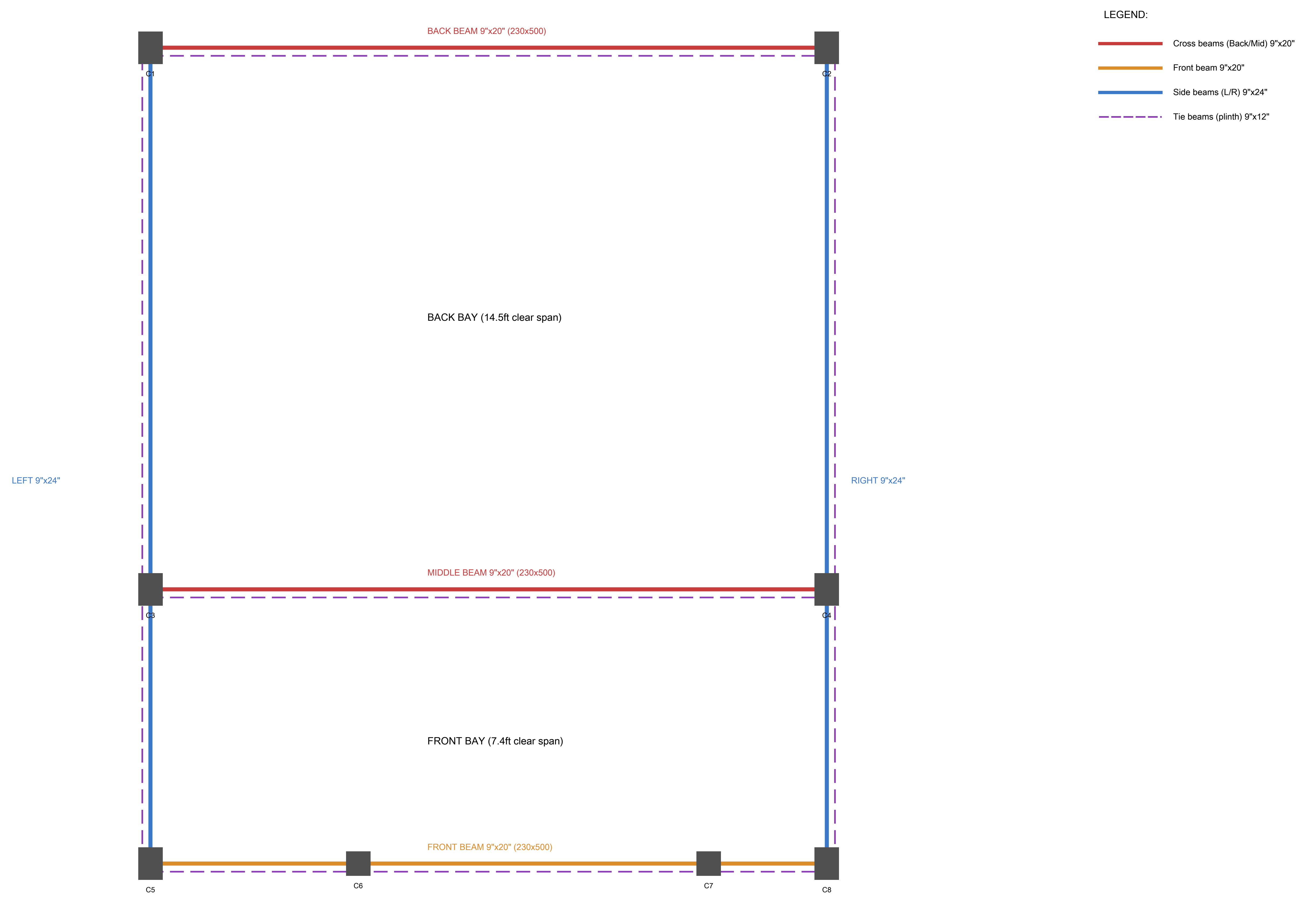
C1 to C5, C8 = 9" x 12" (230x300mm)

C6, C7 = 9" x 9" (230x230mm)

C6, C7 are GF only (do not extend to 1F)

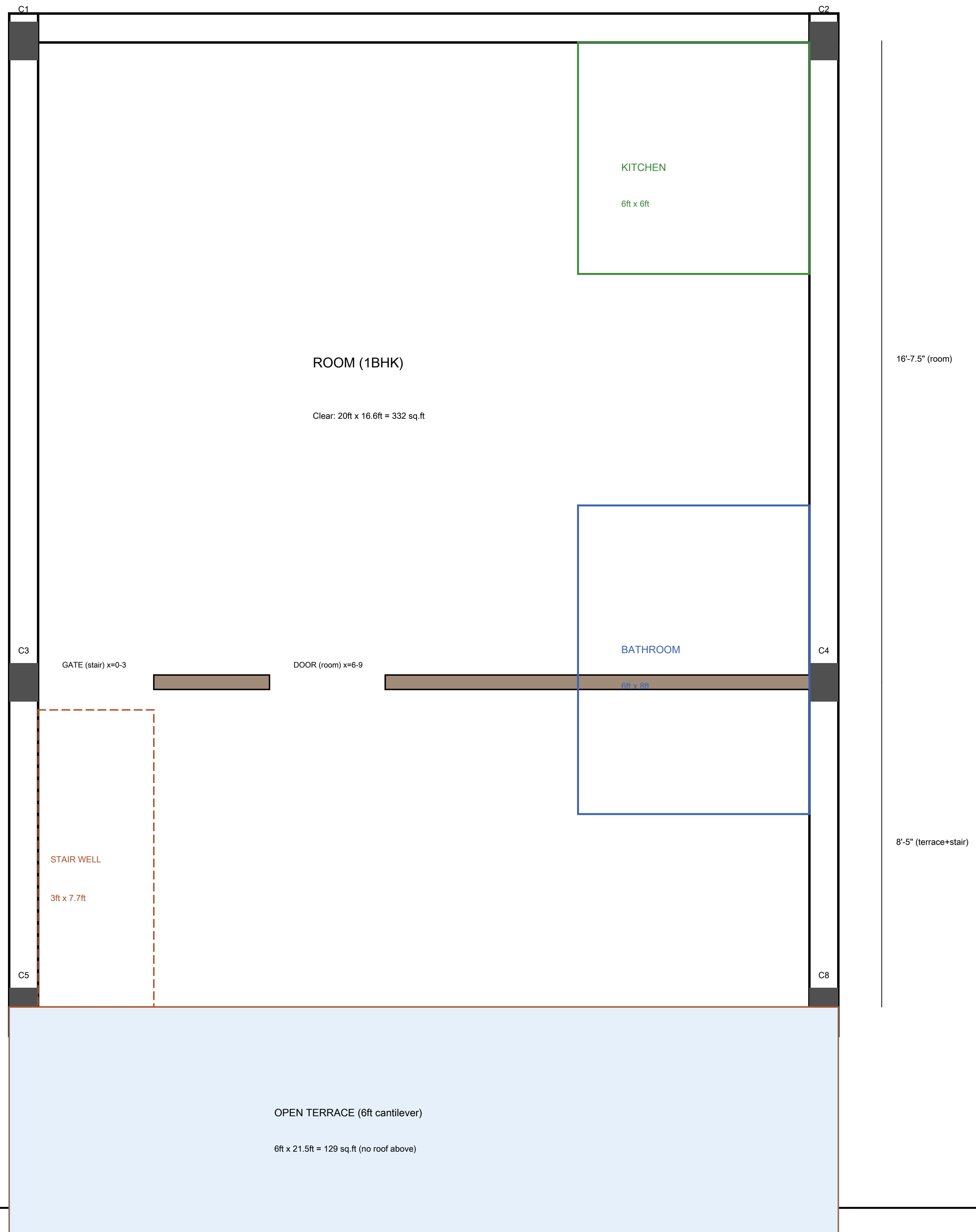
SHEET 3: BEAM LAYOUT PLAN

GF Ceiling Level (z = 15ft). Main beams + Tie beams at plinth level.



SHEET 4: FIRST FLOOR PLAN

Plan at z = 15.5ft (1F floor level). Shows room layout, partition, stairwell, kitchen, bathroom.



1F NOTES:

C6 & C7 do NOT extend to 1F

No front wall (open terrace)

Partition wall: 4.5" brick (115mm)

Floor: 6" RCC slab (from GF beams)

Roof: 6" RCC slab at z=27.5ft

Parapet: 4.5" brick, 3ft high

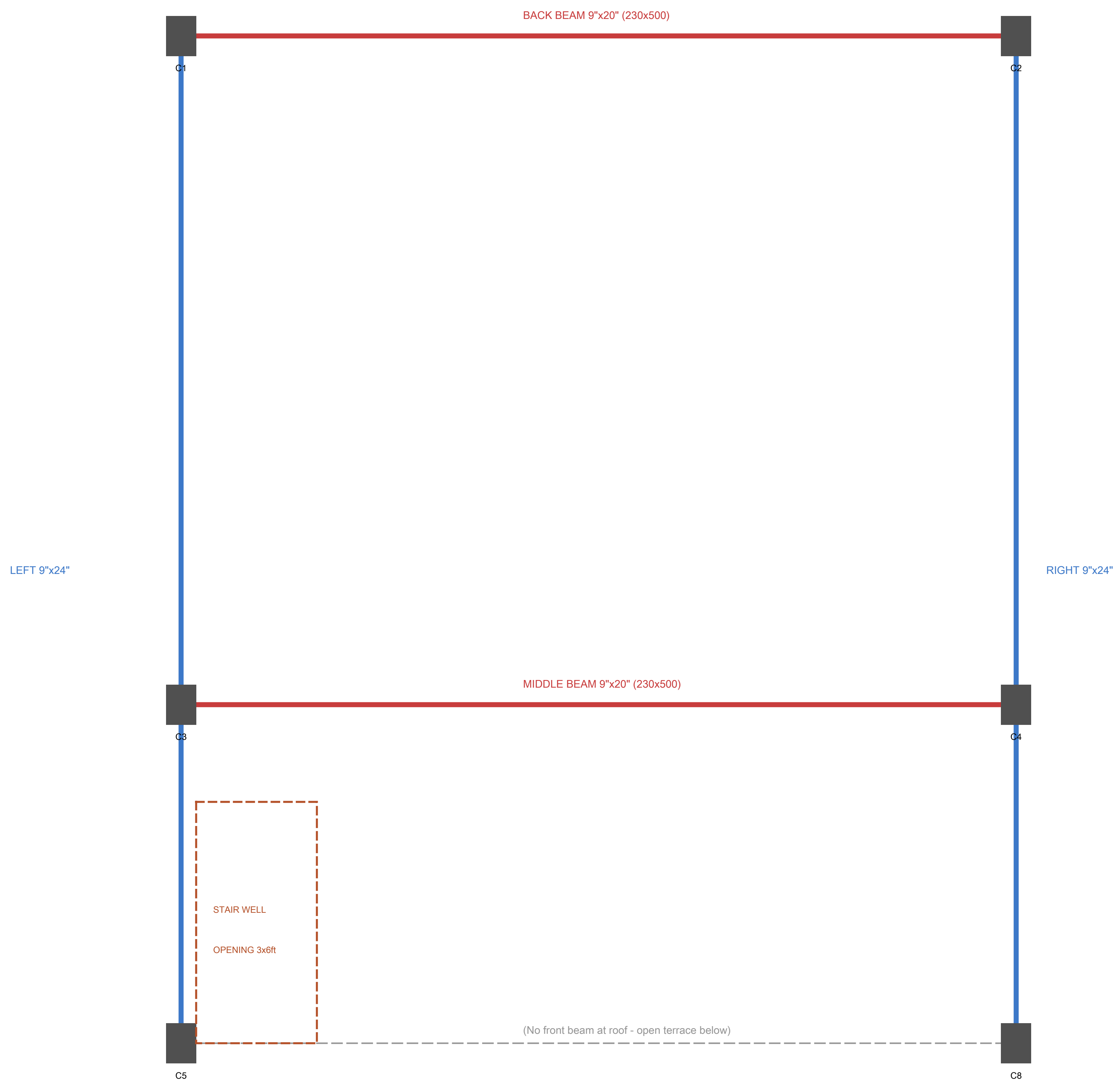
Terrace: cantilever, no roof

Stairwell: 3ft x 7.7ft opening

Sloped stair covers stairwell

SHEET 5: 1F ROOF BEAM LAYOUT (z = 27.5ft)

Same configuration as GF ceiling beams. Beams at roof level supporting roof slab + parapet.



1F ROOF BEAMS:

Same sizes as GF ceiling level beams.

Back + Middle: 9"x20" (230x500mm)

Left + Right: 9"x24" (230x600mm)

No front beam (1F has no front wall)

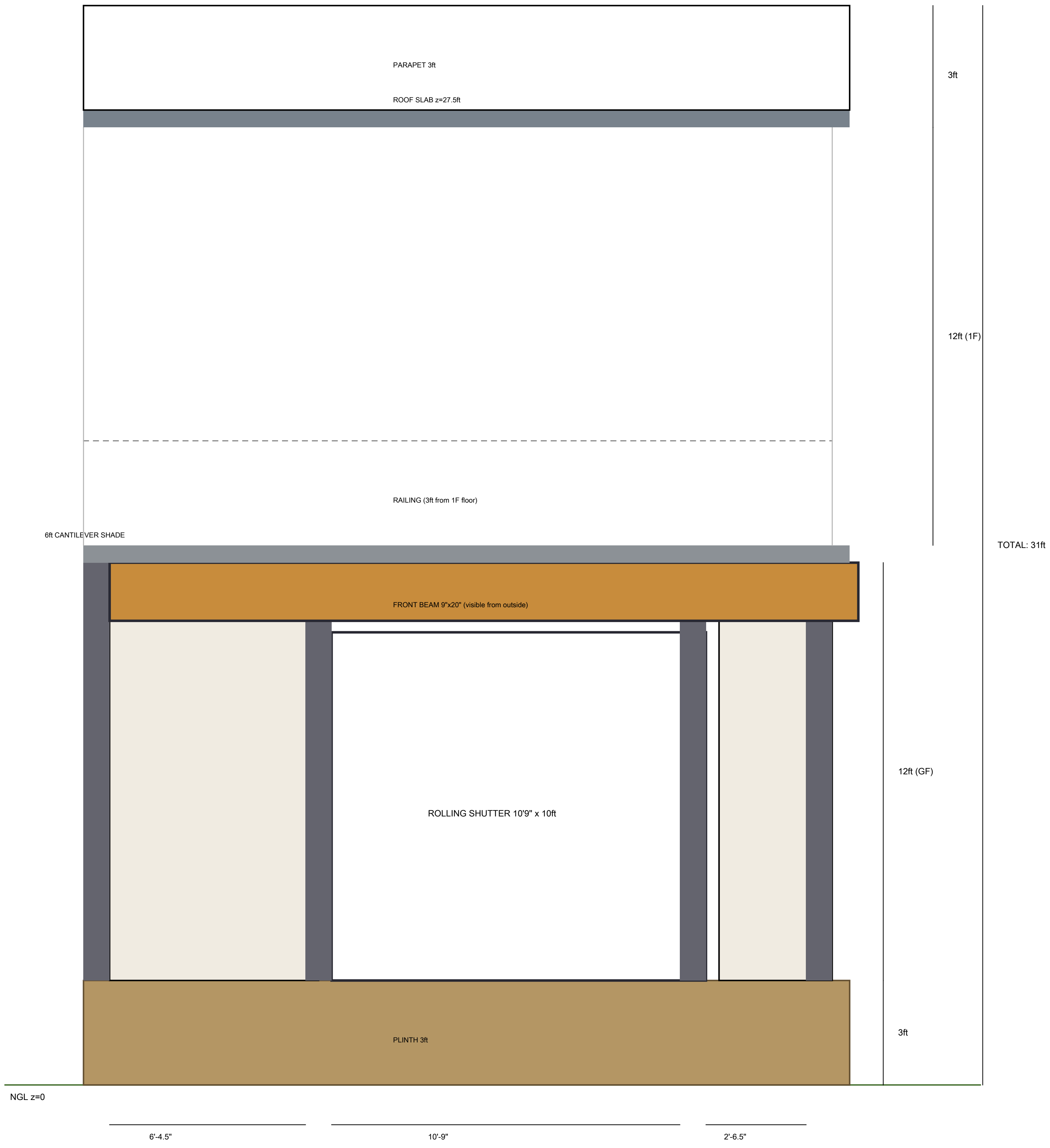
Roof slab: 150mm (6") RCC, same as floor

Parapet: 4.5" brick, 3ft high on all 3 sides

Stairwell opening: 3ft x 6ft (for sloped stair)

SHEET 6: FRONT ELEVATION

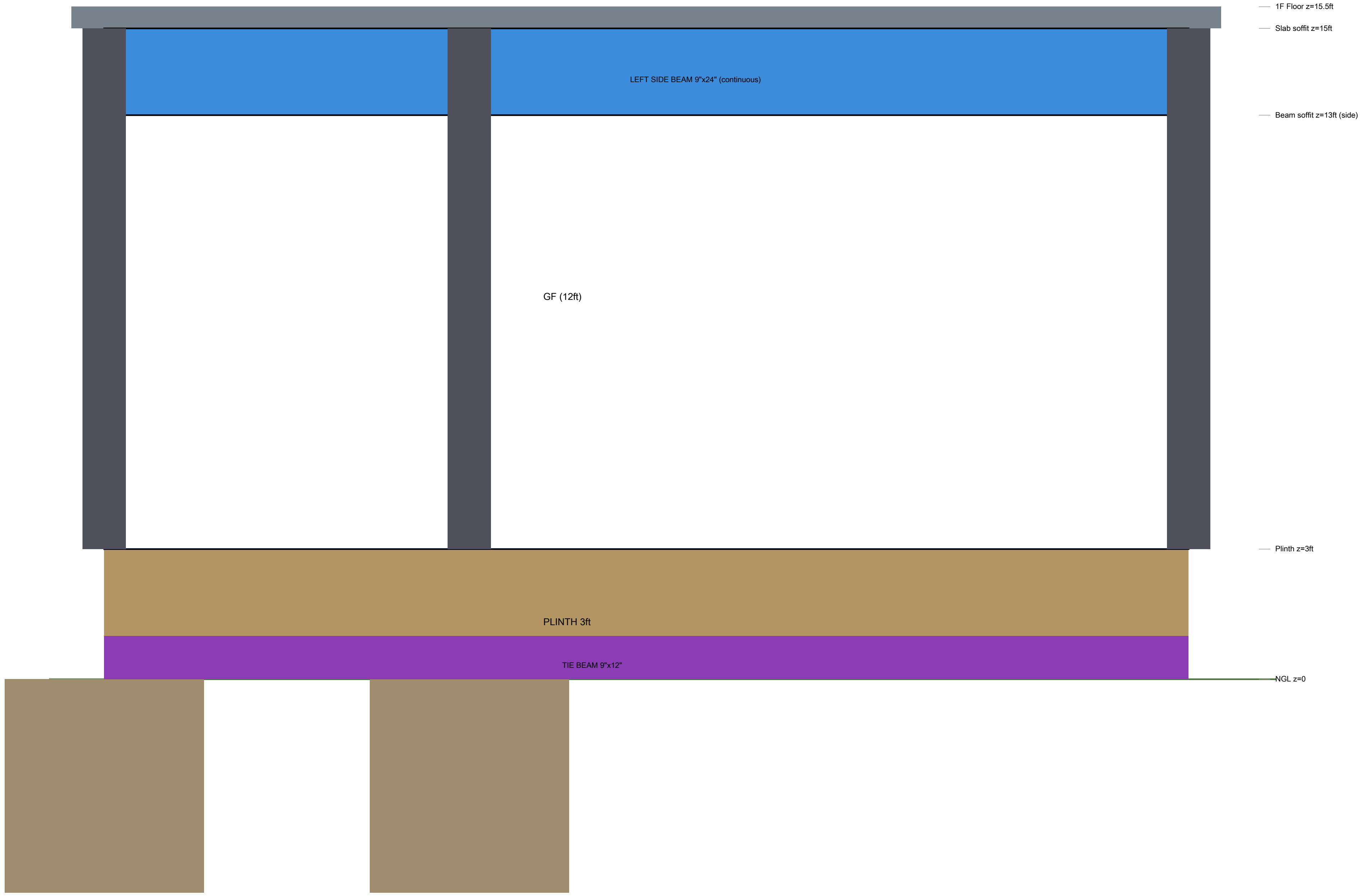
View from OUTSIDE looking at front face. Left = C5 side, Right = C8 side.



SHEET 7: STRUCTURAL SECTIONS

Section A-A (Front to Back, through left wall) | Section B-B (Left to Right, through middle beam)

SECTION A-A (Front to Back)



SHEET 6: COLUMN & TIE BEAM SCHEDULE

A. COLUMN SCHEDULE

Col	Size (mm)	Position	Main Steel	Ties	Height	Util.
C1	230x300 (9"x12")	Back-Left corner	4-16+4-12 (1.82%)	8mm@150/100	GF+1F (24ft)	45%
C2	230x300 (9"x12")	Back-Right corner	4-16+4-12 (1.82%)	8mm@150/100	GF+1F (24ft)	45%
C3	230x300 (9"x12")	Mid-Left (y=8'5")	6-16+2-12 (2.08%)	8mm@150/100	GF+1F (24ft)	52%
C4	230x300 (9"x12")	Mid-Right (y=8'5")	6-16+2-12 (2.08%)	8mm@150/100	GF+1F (24ft)	52%
C5	230x300 (9"x12")	Front-Left corner	4-16+4-12 (1.82%)	8mm@150/100	GF+1F (24ft)	45%
C6	230x230 (9"x9")	Front @ x=6'0"	4-12mm (0.85%)	8mm@150/100	GF only (12ft)	25%
C7	230x230 (9"x9")	Front @ x=16'9"	4-12mm (0.85%)	8mm@150/100	GF only (12ft)	25%
C8	230x300 (9"x12")	Front-Right corner	4-16+4-12 (1.82%)	8mm@150/100	GF+1F (24ft)	45%

B. TIE BEAM SCHEDULE (Plinth Level)

Tie Beam	Size (mm)	Span	Main Steel	Stirrups	Level
TB-Back (C1-C2)	230x300 (9"x12")	20.75ft/6.3m	4-12mm (2T+2B)	8mm @ 200 c/c	Plinth (z=0-1ft)
TB-Middle (C3-C4)	230x300 (9"x12")	20.75ft/6.3m	4-12mm (2T+2B)	8mm @ 200 c/c	Plinth (z=0-1ft)
TB-Front (C5-C8)	230x300 (9"x12")	20.75ft/6.3m	4-12mm (2T+2B)	8mm @ 150 c/c	Plinth (z=0-1ft)
TB-Left (C5-C3-C1)	230x300 (9"x12")	24.75ft/7.5m	6-12mm (3T+3B)	8mm @ 150 c/c	Plinth (z=0-1ft)
TB-Right (C8-C4-C2)	230x300 (9"x12")	24.75ft/7.5m	6-12mm (3T+3B)	8mm @ 150 c/c	Plinth (z=0-1ft)

NOTES:

1. Special confining zone (Lo) = 650mm from each beam-column joint face (IS 13920)
2. Stirrup spacing within Lo = 100mm c/c (8mm dia, rectangular)
3. No lap splices within joint zone. All column laps at MID-HEIGHT only.
4. Lap length = 50d (50 x 16 = 800mm for 16mm bars, 50 x 12 = 600mm for 12mm bars)
5. Column capacity (min steel): 815 kN. Max demand: 363 kN. Factor of safety: 2.25x

SHEET 7: BEAM SCHEDULE (GF Ceiling / 1F Floor Level)

A. MAIN BEAM SCHEDULE

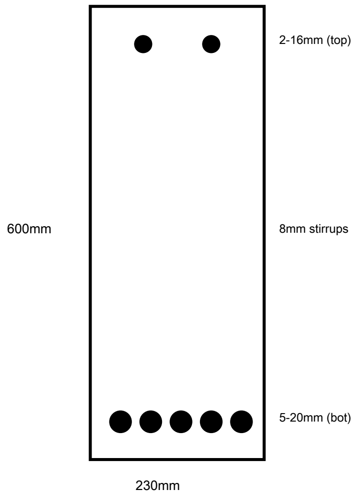
Beam	Size(mm)	Span	Bottom Steel	Top Steel	Stirrups	Mu(kNm)
Back (C1-C2)	230x500 (9"x20")	20.75ft/6.3m	4-20mm+2-16mm	4-20mm+2-16mm	8mm@100/150	266.3
Middle (C3-C4)	230x500 (9"x20")	20.75ft/6.3m	4-20mm+2-16mm	4-20mm+2-16mm	8mm@100/150	266.3
Front (C5-C8)	230x500 (9"x20")	20.75ft/6.3m*	3-20mm+2-16mm	2-20mm+2-16mm	8mm@100/150	143.8
Left (C1-C3-C5)	230x600 (9"x24")	24.75ft/7.5m	5-20mm	2-16+2-20(sup)	8mm@100/175	386.1
Right (C2-C4-C8)	230x600 (9"x24")	24.75ft/7.5m	5-20mm	2-16+2-20(sup)	8mm@100/175	386.1

* Front beam segmented by C6, C7. Max segment span = 10.75ft (C6-C7, shutter opening).

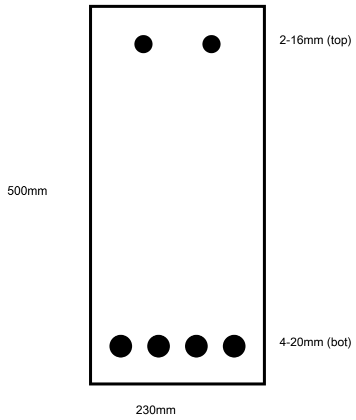
Stirrups: @100mm c/c within L/4 from each support; @150-175mm c/c at midspan.

B. BEAM CROSS-SECTIONS (Typical)

9"x24" SIDE BEAM



9"x20" CROSS BEAM



SHEET 8: SLAB & FOUNDATION DETAILS

A. SLAB SCHEDULE

Slab	Thick.	Type	Bottom Steel	Top Steel	Span
1F Floor (main)	150mm (6")	Two-way	10mm@150 c/c both	8mm@200 at supports	20'x25'/6.1x7.6m
Roof slab	150mm (6")	Two-way	10mm@150 c/c both	8mm@200 at supports	20'x25'/6.1x7.6m
Cantilever (front)	150mm (6")	Cantilever	8mm@200 (bottom)	12mm@125 (TOP-main)	6ft/1.83m proj.
GF floor (on earth)	100mm (4")	On grade	6mm mesh@150	N/A	On soil
Staircase waist	125mm (5")	One-way	12mm@150 c/c	8mm@200 (distr.)	~8ft/2.5m

B. FOUNDATION SCHEDULE

Footing For	Size	Depth	Thickness	Steel	SBC	Type
Interior (C3,C4)	1.4x1.4m (4.6'x4.6')	1.5m/5ft BGL	300mm/12"	12mm@150 both	130 kN/m2	Isolated
Corner (C1,C2,C5,C8)	1.4x1.5m (4.6'x5')	1.5m/5ft BGL	300mm/12"	12mm@150 both	130 kN/m2	Isolated
Front (C6,C7)	1.2x1.2m (4'x4')	1.5m/5ft BGL	250mm/10"	10mm@150 both	130 kN/m2	Isolated

All footings connected by tie beams (230x300mm) at plinth level. PCC bedding: 150mm M10 below all footings.

SHEET 9: GENERAL CONSTRUCTION NOTES & SPECIFICATIONS

A. सामग्री (MATERIALS - मुरादाबाद)

सीमेंट: M20 (1:1.5:3) - ACC/Ambuja/Ultratech OPC-43. कम से कम 7 बैग/m3.

सरयिा: Fe500D TMT - TATA Tiscon / Kamdhenu 500D / JSW (BIS मार्क ज़रूरी).

ईंट: पक्की ईंट, पहली श्रेणी - स्थानीय भट्टा (7.5 N/mm2+).

बालू: Zone-II साफ नदी रेत - गंगा/रामगंगा। मट्टिी-रहति।

रोड़ी: 20mm कुचली पत्थर - हरद्विार/रुड़की/नजीबाबाद।

पानी: पीने योग्य साफ पानी। नमकीन/गंदा पानी नहीं।

B. कवर (COVER - सरयि की दूरी)

नींव (Foundation): 2 इंच (50mm) नीचे, 1.5 इंच (40mm) साइड

पलिर (Column): 1.5 इंच (40mm) चारों तरफ

बीम (Beam): 1 इंच (25mm) नीचे और साइड

छत/स्लैब (Slab): 3/4 इंच (20mm) नीचे, 5/8 इंच (15mm) ऊपर

कवर कम होने से सरयिा जंग खाएगा। कभी कम न करें!

C. ओवरलैप/गोद (LAP LENGTH - Zone IV)

खचािव (Tension): 50 x व्यास। 20mm = 1000mm (3 फुट 4 इंच).

दबाव (Compression): 40 x व्यास। 16mm = 640mm (2 फुट 1 इंच).

पलिर में गोद: सर्फि बीच में (6 फुट ऊंचाई)। जोड़ के पास कभी नहीं!

छज्जे की ऊपरी सरयिा: मुख्य स्लैब में 10 फुट अंदर तक जाएगी।

गोद पर 2 बार बाँधना (binding wire double)।

D. भूकंप सुरक्षा (SEISMIC - IS 13920)

पलिर-बीम जोड़ (Joint) में रगि: 4 इंच (100mm) c/c - बहुत करीब!

जोड़ से 2 फुट 2 इंच (650mm) तक करीबी रगि (Lo zone)।

रगि का हुक: 135 डग्रिी मोड़। 90 डग्रिी हुक मना है!

बीम की नीचे की सरयिा: कम से कम ऊपर की आधी जोड़ से पार जाएगी।

मुरादाबाद Zone-IV - भूकंप का खतरा ज़्यादा - डटिलगि ज़रूरी!

E. ढलाई/देखभाल (CURING & SEQUENCE)

क्योरगि: 14 दनि लगातार पानी (छत: ponding, पलिर: गीली बोरी)।

शटरगि हटाना: बीम साइड 3 दनि, स्लैब प्रॉप 21 दनि, छज्जा 28 दनि।

क्रम: नींव > प्लथि बीम > भराई > फ्लोर > पलिर > बीम+स्लैब (साथ में).

ढलाई 30 मनिट में पूरी करें। बाद में पानी मलिाना मना है।

वाइब्रेटर ज़रूरी (सुई वाला)। हाथ से ठोकना काफी नहीं।

गर्मी में सुबह/शाम ढलाई। सर्दी में बोरी/पॉलीथीन से ढकें।

SHEET 10: INDEPENDENT STRUCTURAL REVIEW & RECOMMENDATIONS

A. OVERALL STRUCTURAL CONFIDENCE ASSESSMENT

This structure has been reviewed against IS 456:2000, IS 1893:2016, and IS 13920:2016 for a G+1 RCC frame building in Seismic Zone IV. The 8-column moment-resisting frame (OMRF, R=3.0) with as-built column positions is assessed for gravity loads, seismic forces, and serviceability.

B. ELEMENT-WISE ADEQUACY RATING

Element	Rating	Basis	Concern (if any)
Columns (230x300)	ADEQUATE	45-52% utilization, FoS 2.25	Strong col-weak beam marginal
Side beams (230x600)	ADEQUATE	Mu,cap > Mu,demand (doubly reinf.)	At design limit for 7.62m span
Cross beams (230x500)	ADEQUATE	Doubly reinforced design	Heavy at 1F level (266 kNm)
Front beam (230x500)	ADEQUATE	Segmented by C6/C7	Shutter span unsupported below
Tie beams (230x300)	ADEQUATE	Nominal ties for seismic	L/R tie beam: 15.5ft span is long
Floor slab (150mm)	MARGINAL	OK for strength	Back bay deflection borderline
Cantilever (150mm, 6ft)	MARGINAL	Strength OK	L/d ratio 14.6 > 9.8 limit
Foundations (1.4x1.4m)	ADEQUATE	SBC utilization ~60%	Needs soil test confirmation
Seismic drift (Y-dir)	MARGINAL	Amplified drift ~39mm	Exceeds limit; infill helps

C. RECOMMENDATIONS FOR IMPROVEMENT (Without Major Rework)

1. OPTIONAL PILLAR at x=7'1.5", y=8'5" (near middle beam, left of C6): Provides direct load path for staircase reactions into the middle beam. Reduces beam bending at that point. Size: 9"x9" (230x230mm), 4-12mm steel, same height as GF columns. Low cost, high benefit.

2. CANTILEVER EDGE BEAM: Add a 150x230mm upstand beam at the 6ft cantilever tip. Reduces deflection by ~40%. Alternatively, increase cantilever slab to 175mm. Cost: ~Rs 3,000 additional. Recommended for long-term crack prevention.

3. BACK BAY SLAB: Increase from 150mm to 175mm (7") for the back bay panel only (between middle beam and back beam). Resolves deflection concern (L/d improves to 28). Cost: ~Rs 12,000 additional concrete + steel. Strongly recommended.

4. INFILL MASONRY: The 9" brick walls (though non-structural) will significantly improve lateral stiffness and reduce seismic drift to well within limits. Ensure proper separation gaps (10-15mm) at beam-wall interface to prevent strut damage. OR mortar-fill for added stiffness.

5. SIDE BEAM MID-SPAN: If any deflection visible during construction, add MS angle bracket (100x100x10mm) as a knee brace from C3/C4 column face to beam soffit at mid-span.

D. PROFESSIONAL SIGN-OFF

STRUCTURAL ADEQUACY CERTIFICATE

Based on preliminary structural analysis per IS 456:2000, IS 875:1987/2015, IS 1893:2016, and IS 13920:2016, this G+1 RCC frame structure is found ADEQUATE for gravity and seismic loads with the noted recommendations. The structure is APPROVED FOR CONSTRUCTION subject to: (a) implementation of recommendations #2 and #3 above, (b) proper seismic detailing per IS 13920, and (c) final design verification by a licensed structural engineer before beam/slab casting.

Confidence Level: 85% (GOOD)   Risk Category: LOW-MODERATE